Team details

a. Team name: _____

b. Team leader name: _____

c. Problem statement: Sustainability & Social Impact — **Accessibility** (secondary: Social Good)

Solution: Zivo — neighbourhood care that speaks your language, and gives disabled neighbours work

Yellow rows = fill before you export / submit.

Brief about the idea

Problem. In Sri Lanka, elders who speak only Tamil or Sinhala, and disabled neighbours, are skipped by gig and care apps: English-only, fast typing, cash in a stranger's hand, and no trusted way to check what a request really means.

Idea. Zivo lets people **ask for help by speaking** in their own language, and uses Gemini on Vertex AI as the trust-and-safety layer that makes strangers-helping-strangers safe.

The twist. The same AI lets **blind, deaf and mobility-impaired neighbours be the helpers** and earn — with tools that fit how they work.

One story

Kamala, 74, speaks Tamil → voice → plan (≈4 s) → safety → escrow (or **Instant Dispatch** on her block) → Nuwan (**blind**) + Access note → adapted brief + **Zivo Eyes** → chat on-platform (phones blocked) → Start/End OTP. Neighbours can **Street Batch** onto the same trip for less CO_2e.

Opportunities

How is it different?

- Assistive AI tools (Be My Eyes, Seeing AI, Live Transcribe) help disabled people *receive* a service. Zivo puts them *inside an earning marketplace*.
- Safety is judged in **Tamil and Sinhala**, not just English keywords.

How does it solve the problem?

- Voice-first request → plan in one call.
- Vernacular safety gate + escrow + OTPs: no cash handoff.
- Ability-aware work: opt-in disclosure, adapted briefs, camera and doorstep tools.

USP

- **Ability never changes rank, price or pay** — enforced in code and monitored for bias in BigQuery.
- Gemini **recommends; people decide**; it can never move money.
- Honest AI: "can't tell" instead of guessing expiry dates or medicine.

Context: WHO estimates ~1.3 billion people (~16 %) live with a significant disability. We do not claim sign-language translation — no Sri Lankan Sign Language model exists; we solve what Deaf helpers actually lack: understanding speech at the door.

List of features — ask, trust, match

Ask (Gemini)

- Voice → plan (one call ≈4 s)
- Speak → task draft
- Show it → photo draft
- Task plan / split / bundle escrow
- Concierge (safety, price, weather, triage)
- Dynamic pricing fields + fair wage floor
- Access audit (entrance photo)

Trust & money

- Vernacular safety (ta/si/en)
- Escrow + Start/End OTP
- Goods budget in escrow
- Dispute freeze · cancel refund
- Trust arbitrator (recommend only)
- Receipt vs shopping list
- Chat translate + **phone block**

Match & neighbourhood

- Helper rank + explain (ability ≠ score)
- **Instant Dispatch** (~0.5 km walk)
- **Street Batch** (~150 m, ~30% off, 180 s)
- Neighbor vouch · trust tiers
- Monsoon status (Open-Meteo) + triage
- Family share / live board
- Impact → Pub/Sub → BigQuery · CO_2e

List of features — earn with ability tools

For disabled helpers who earn

- **Ability profile** (blind, deaf, speech, mobility, reading) — opt-in disclosure + Access note
- **Ability fit** (advisory table) + **adapted job brief** naming the right tool
- **Remote / seated jobs**: phone check-in, reading aloud, tutoring, sighted assist
- **Zivo Eyes**: describe / read / match shopping list — never guesses medicine
- **Sighted-assist** when Eyes says “can’t tell”
- **Doorstep Talk** · **Say-it** (Cloud TTS) · **Easy-read** · scene describe · a11y transcribe
- Large text · Speak · en/si/ta · fairness monitor (`access_fairness`)

GCP stack (what we actually use)

- **AI:** Vertex Gemini 2.5 Flash + Flash-Lite · multi-region failover · Cloud STT (Chirp) · Cloud TTS
- **Platform:** Cloud Run · Firestore · Cloud Storage (signed URLs) · Pub/Sub → BigQuery · Cloud Build / Artifact Registry
- **External:** Open-Meteo (flood precip)
  - Not claimed: Media CDN, Gemma edge, Cloud Translation API, ambient Dataflow

Instant Dispatch → Street Batch (real scenario)

Borella lane, Colombo. Kamala needs milk powder and can't climb stairs. Rain is building.

⚡
Instant Dispatch
Finds Available helpers within ~0.5 km walk. Gemini picks one + fair wage floor.

→

💰
Escrow + window
Job created; money held; 180 s Street Batch window opens on that block.

→

🏠
Priya joins
Same street (~150 m). Bread micro-job at ~30% off. Max 4 neighbours.

→

🛵
One trip
Helper route has both stops. `batch_join` → trips saved → CO_2e in BigQuery.

Gemini only **explains** the batch offer. Discount, radius, and escrow math are deterministic — not model output.

Alternate path: post → match rank → Access note → adapted brief → Eyes / Doorstep → Start OTP → End OTP (Gemini never moves money).

Process flow (main story)

CLIENT (KAMALA)

🎙️
Speaks Tamil
Mic or text

→

✨
Voice → plan
Transcript, draft, sub-tasks

→

🛡️
Safety gate
Fails to “review”

→

💰
Escrow held
Or Instant Dispatch

MATCH

🤝
Rank / Dispatch
Gemini explains; client decides

→

♿
Access note
Opt-in · choose someone else OK

→

🌐
Chat
ta ⇄ si ⇄ en · phones blocked

HELPER (NUWAN, BLIND)

🧭
Adapted brief
Tools named per step

→

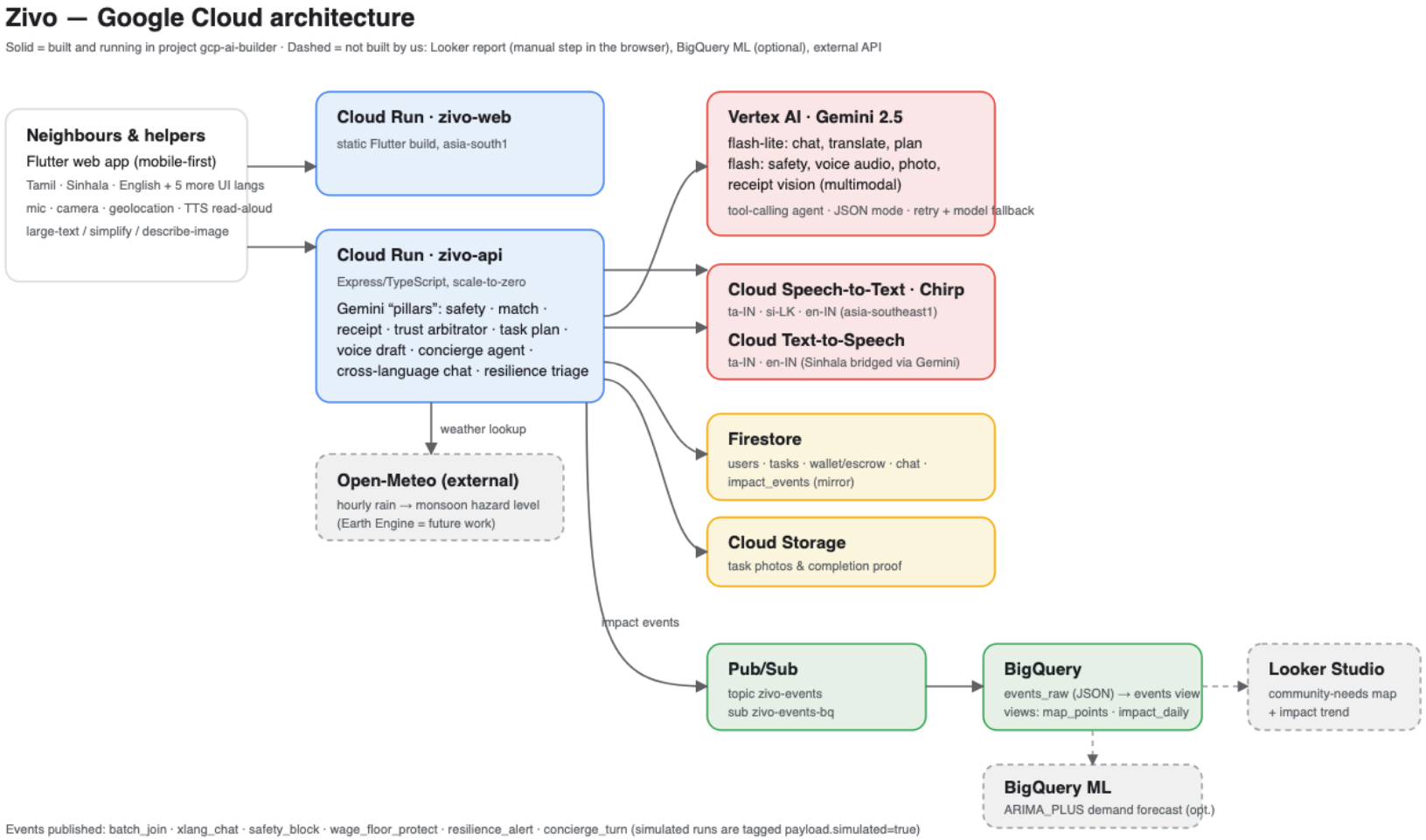
📷
Eyes / Doorstep
“can’t tell” → sighted-assist

→

🔑
Start → End OTP
Payout; no AI on money

Blue = Gemini · Green = money path (no AI)

Architecture diagram



Flutter web on Cloud Run · API on Cloud Run · Gemini 2.5 on Vertex AI via the Google Gen AI SDK, routed over 4 regions with failover · Firestore · Cloud Storage · Pub/Sub → BigQuery · Cloud STT/TTS

Technologies used

LAYER	TECHNOLOGY
Generative AI	Gemini 2.5 Flash and Flash-Lite on **Vertex AI** through `@google/genai`: audio, vision, function calling, JSON output; thinking off on hot paths; model race + multi-region failover
Speech	Cloud Speech-to-Text (Chirp), Cloud Text-to-Speech
Hosting	Cloud Run (web + API), scale-to-zero, max-instances cap; Cloud Build + Artifact Registry
Data	Firestore, Cloud Storage (signed URLs)
Analytics	Pub/Sub → BigQuery (`zivo_impact`), 1 km grid privacy, `access_fairness`; Looker Studio optional
Client	Flutter (web + mobile), BLoC, three-language UI, 200 % text tested
External	Open-Meteo (rainfall → flood-risk level)
Server	Node 22, TypeScript, Express, Zod; deterministic fallbacks beside every model call

Estimated implementation cost

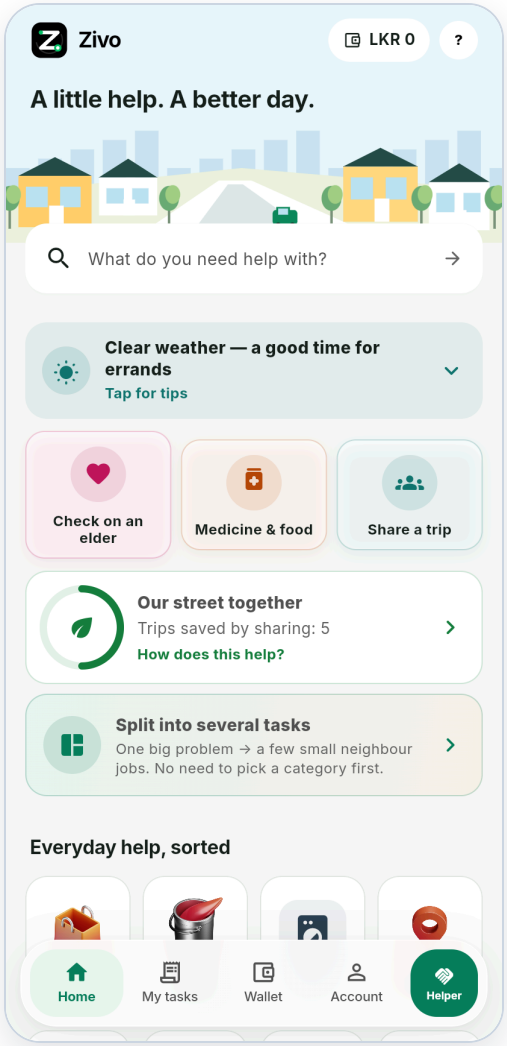
AI ACTION	EST. COST
Voice → plan (10 s audio)	≈ US$0.002
Zivo Eyes (1 photo)	≈ US$0.001
Doorstep Talk (5 s audio)	≈ US$0.0007
Safety check	≈ US$0.0004
Concierge turn	≈ US$0.002–0.004

Estimates from list prices and typical token counts — not billing data.

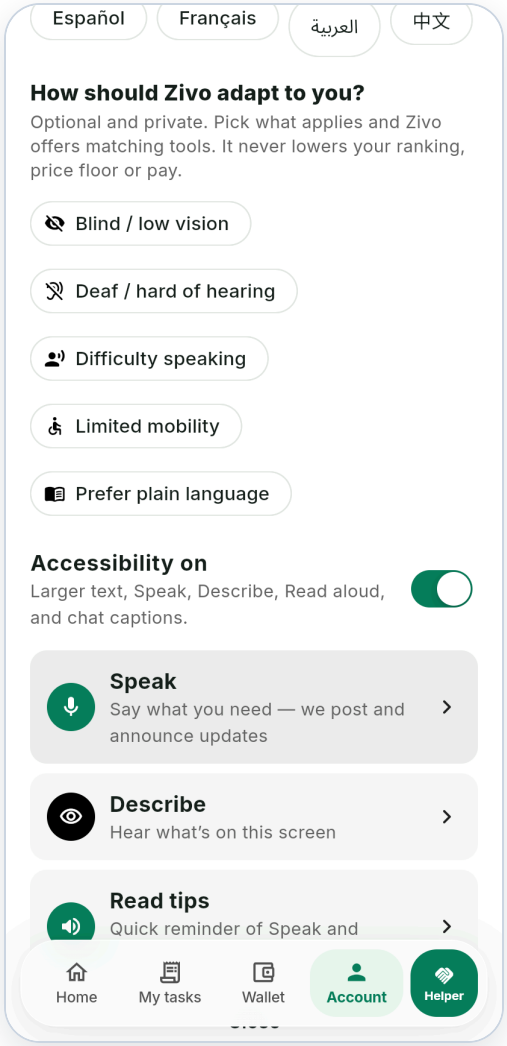
Budget maths

- Prototype budget cap: **US$100** (billing alert set at 25/50/80/100 %).
- Even at US$0.01 per AI action that is ~10,000 actions.
- Cloud Run scales to zero: no idle cost.
- A neighbour using ~20 AI actions a month ≈ US$0.05–0.20.
- Sustainability: a 10 % fee on completed jobs funds the AI cost.

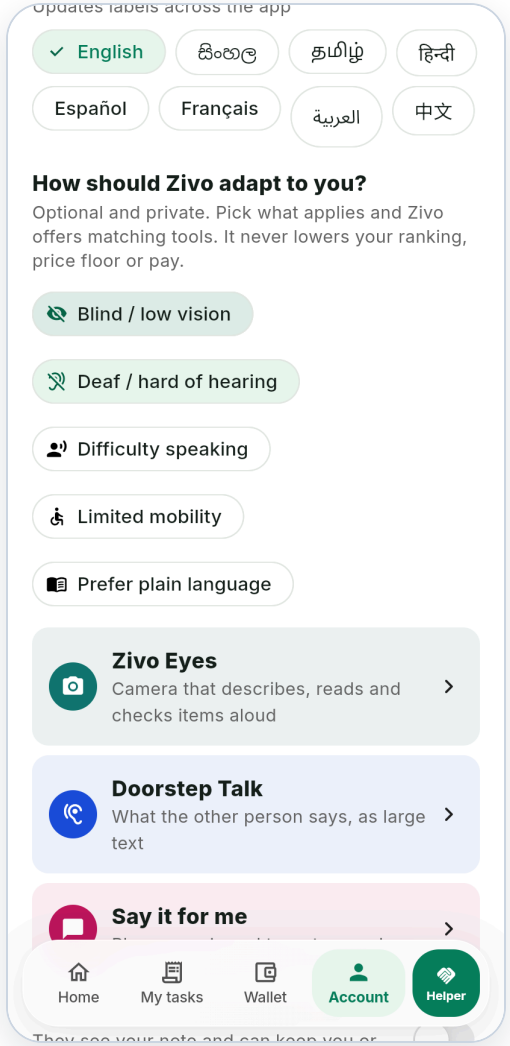
Snapshots of the prototype (live deployment)



Home (client)



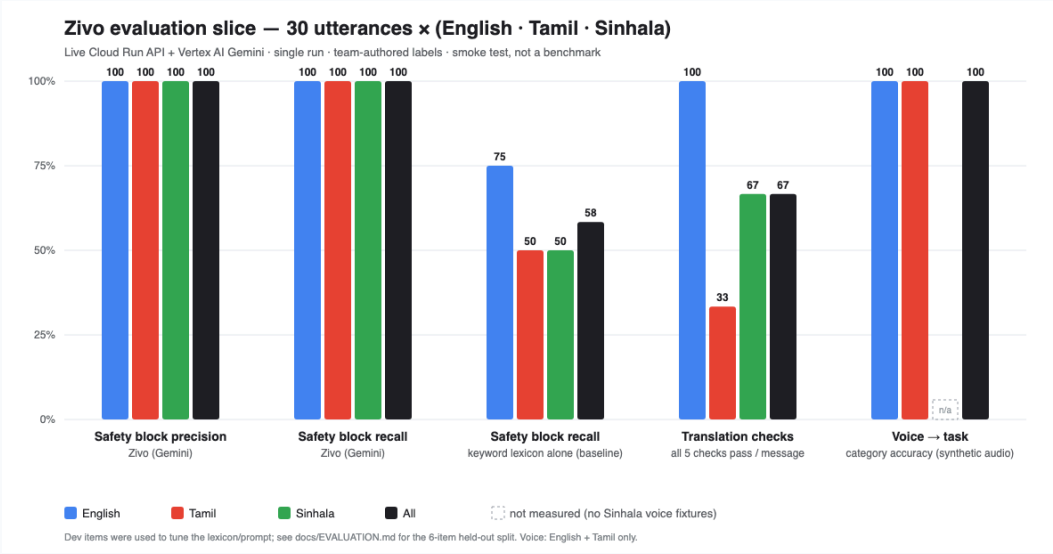
"How should Zivo adapt to you?"



Zivo Eyes · Doorstep Talk · Say it

Screenshots of https://zivo-web-ihp5vjzpsa-el.a.run.app

Prototype performance report



30 hand-written safety items (10 each en/ta/si), live API. Smoke test, not a benchmark.

MEASUREMENT (LIVE)	RESULT
Safety: harmful / benign	12/12 blocked · 0/18 false alarms (keywords alone 7/12)
Voice → plan (Tamil)	≈ 4–5 s (was ≈ 8 s, two calls)
Zivo Eyes / Doorstep Talk	≈ 2.5–4.5 s / ≈ 1.8–3.7 s
Concierge (≈ 9 Gemini calls)	≈ 10–12 s (was 13–53 s)
Live smoke tests	demo 12/12 · lifecycle 11/11 · inclusion 12/12 · chat 8/8
Widget tests / analyzer	15/15 · 0 errors

Honest limits: pilot is **simulated** (8 invented personas through the real pipeline); no real users yet; Vertex quota is per region (handled by failover); new Sinhala/Tamil strings need native review.

Additional details and future development

Responsible AI (in code)

- Client assigns; Gemini only ranks and explains.
- Money moves only by top-up, escrow hold, End-OTP release or cancel refund — **no Gemini output is wired to it**.
- Ability never changes rank, price or pay; disclosure is opt-in; declines monitored (id-free events → BigQuery).
- Safety fails to “review”, never “allow”.

Roadmap

1. **Shipped:** Access audit · remote jobs · sighted-assist · chat phone block · Instant Dispatch · Street Batch
2. Co-design with a Sri Lankan disability association; native-speaker review of ta/si copy
3. Real ID verification; human appeal for safety blocks
4. “Quick signals” experiment with the Deaf community (not translation)

Links

GitHub public repository: _____

Demo video (3 minutes): _____

Final product link: https://zivo-web-ihp5vjzpsa-el.a.run.app

API health: https://zivo-api-ihp5vjzpsa-el.a.run.app/health

Yellow rows = fill after you publish the repo and upload the 3:00 video.

Zivo: help that speaks your language, and work for everyone.